

Containment is the wrong north star.

The industry scores tickets closed without a human, so agents learn to act confidently rather than hand over.

WHAT I BUILT TOWARD INSTEAD

Resolution without harm, at a price that scales.

The agent proves it knows enough to act *before* it acts. The model understands the customer; code decides. The seam is the design.

	SCRIPTED BOT	THIS AGENT	AUTONOMOUS AGENT
understands	menus and keywords	the model	the model
decides	code	code	the model, by prompt
hands over	when the tree runs out	by rule	when it decides to

THE SHAPE IS NOT NEW

Automated decisioning, as banks and insurers run it: decide alone where risk is low, bring a person in where it is not.

```
risk score          -> triage.py
rules engine        -> policy.py
manual review queue -> escalate_to_human
case ownership      -> after handover, no model call
```

The Tahoe for a dollar and the invented bereavement fare are the tolerance this is built for.

Screen cheaply, gate hard, decide deterministically.

REQUEST PATH

triage.py	Haiku routes. Mistral screens abuse and self-harm.
agent.py	Direct SDK, bounded tool loop, no framework.
tools.py	Eight tools. Every refusal is in the result.
policy.py	Eligibility as code. The model never evaluates it.
grounding.py	Prose held to the published text.
providers.py	One env var pins to eu.openrouter.ai.

Memory is the conversation on the Agent instance. Nothing hidden. **The prompt** is 300 words of tone and judgement with no rule in it.

FOUR GATES, ALL STRUCTURAL

```
start_return(order_id, reason)
    ^ required, and never inferred

find_orders(email) -> every match

    2 matches -> nothing to act on
                -> the model must ask
```

Prompting "ask when unsure" usually happens. Removing the code path always does.

Which order. Whether policy allows it. Whose account, by a code the model never sees. Who owns the conversation: after a handover, no model call.

TRADED AWAY, AND WHY IT WAS WORTH IT

A turn to ask which, a turn to verify. The wrong Dune is a coin flip the customer sees, and an email address is not an identity.

Don't prompt what should have been code.

A prompt is advisory. A function returns the same answer whether the customer is polite or furious, and it is testable.

```
age = (today - order.delivered_on).days
if age > RETURN_WINDOW_DAYS:
    return Decision(
        False, "outside_window",
        f"Delivered {age} days ago, outside "
        f"the {RETURN_WINDOW_DAYS}-day window.")
```

How the code gets written does not matter. What runs at decision time does.

Prose gets the same rule: a second model holds every general answer to the published text. Explain the policy; never extend it.

EVERY REFUSAL HAS A CODE

ORDER	VERDICT	REASON
BK-10231	allow	Delivered 5 days ago
BK-09988	deny	outside_window
BK-10250	deny	digital_item
BK-10244	deny	not_yet_delivered
BK-10231 "never came"	deny	delivery_dispute

TRADED AWAY, AND WHY IT WAS WORTH IT

Flexibility: no goodwill path, that lives behind a person. In return, every refund has the same answer for every customer, and a test.

Opus to read a database row is not a business.

Haiku picks the tier. Mistral, free, screens abuse and self-harm. Haiku keeps bookstore fraud.

	PER CONVERSATION	10K/DAY	PER YEAR
routed	\$0.0167	\$167	\$61,039
all Opus	\$0.0596	\$596	\$217,670
saved	72%		\$156,631

209 real calls, list prices. 3 of 24 conversations went to Opus: the fraud signal, the threat, a planted instruction.

Swap the model and the destinations hold. DeepSeek V4.1 Flash on both tiers: 20 of 20, \$0.0012 a conversation.

WHY A MISROUTED TURN IS SURVIVABLE

```
"I'd like to return my Dune"
  triage -> simple -> haiku-4-5
  find_orders -> 2 matches
  agent asks which one           OK
```

Routing is an optimisation; correctness never depends on it. A triage miss costs a less polished reply and never a wrong refund.

THE SECOND JOB

A threat is logged and the eligible refund still goes through; tone is not policy. A self-harm signal gets fixed text, crisis resources and a person. No model answers it.

TRADED AWAY, AND WHY IT WAS WORTH IT

A less polished reply on a misrouted turn, and abusive customers cost more because risk routes heavy. In return, 72% off the bill with no wrong refund in 24 cases and 40 phrasings.

14 of 24 pass only by refusing, asking, holding, verifying, or not inventing.

CASES	ASSERT	
Refuses on a policy code (5) outside_window · digital_item · not_yet_delivered · never_received · bypass_pressure	no refund; the code is recorded	5/5
Asks or holds (5) ambiguous_order · no_identifier · name_only · someone_elses_order · after_handover	nothing guessed, nothing acted on	5/5
Acts when it should (6) eligible_return · title_plus_qualifier · typo_in_id · threat_still_served · asks_for_human · self_harm_signal	refund or escalation fires	6/6
Verifies before reading (2) wrong_code · order_number_alone	nothing read without a verified email	2/2
Prose held to the published text (6) password_reset · shipping_time · student_discount · my_order · follow_up · injected_instruction	looked up, checked, replaced if extended	6/6

24 / 24

40 of 40 phrasings, terse to furious, same outcome.
Every eligibility decision carries a policy code.
0 account reads before verification.
0 model calls after a handover.

WHAT I WOULD CHANGE FIRST, GIVEN A PRODUCTION CONTEXT

Build the eval set from real transcripts. The two holes that would have cost money were found by a person typing at the UI while every case was green. Invented cases only contain what you already thought of.